

UQ-AAS21: A Comprehensive Dataset of Amazon Alexa Skills

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Abstract. Various virtual personal assistant (VPA) services have become popular, due to the convenient interaction manner of voice user interface (VUI) they offer. Centered around them, an ecosystem involving service providers, third-party developers and end users, has started being formulated. The developers are enabled to create applications and release them through application stores, from which the users can obtain them and then run them on smart devices. This emerging ecosystem is still in its early stage, and a great deal of research effort is desired to make it on the healthy track to facilitate its development. Nonetheless, there is still a lack of comprehensive datasets for our research community to conduct research on relevant issues, e.g., the bug-freeness and quality of the applications, and users' security and privacy concerns on them. In this work, we aim to build such a dataset for research use. We target the Amazon VPA service, i.e., the Alexa, which is the most popular VPA service. We collect 65,195 Alexa applications (or *skills*), and extract comprehensive information about them, including invocation names, user reviews, among overall 16 attributes. We show the demographic details of the skills and their developers, and also conduct preliminary statistical analyses on the quality and privacy issues, to demonstrate the potential usage of our dataset. The dataset and analysis results are released online to facilitate future research: <https://github.com/xie00059/Amazon-Alexa-UQ-AAS21-datasets>.

Keywords: Alexa Skills · Voice user interface (VUI) · Dataset.

1 Introduction

The boom of the Internet of Things (IoT) in recent years has accelerated the integration of various smart devices into our daily life. They are reshaping the traditional way of graphics-based human computer interaction into the voice-based user interaction. Various AI (artificial intelligence)-backed virtual personal assistant (VPA) services are easily accessible on the smart devices, and serve people in a convenient interaction manner of voice user interface (VUI). They handle a wide range of user requests, from making phone calls, setting alarms, playing music, looking up weather forecasts, to controlling smart home devices. According to Statistics [4], VPAs have been deployed on billions of devices around the world, and this number is estimated to reach 8.4 billion by 2024.

Besides the embedded capabilities, the VPA services enable third-party developers to provide applications as an enrichment of their functionalities. The developers create applications and publish them through application stores. The users can then activate an application verbally on their smart speakers and interact with it through oral commands, in place of the traditional mouse movement or screen tapping. Through this model, an ecosystem similar to the Android one that is centered around the Google play market [6], is gradually formulated. Take the Amazon’s VPA service, i.e., Alexa, as an example. The number of its applications has gone within three years from 130 to over 100,000 around all its stores as of September 2019 [8]. As another example, the number of applications for Google Assistant, Google’s VPA service, has risen 2.5 times within 2018 [16].

Due to this rapid growth, the VPA ecosystem has also attracted much attention from various research communities. Some studies have been conducted to analyze VPA users’ behaviors [10, 20, 24], and security implications [23, 29] and privacy concerns [15, 18, 22] of the VPA services. When a dataset is needed, the researchers tend to crawl one on their own. This is however a burdensome and time-consuming process. In addition, the dataset maintenance is troublesome. If an application, or some field of an application, e.g., a piece of user review, is not well kept, it may have become unavailable online when it is needed. Therefore, a comprehensive, well-organized, and publicly available dataset is desirable.

In this work, we aim to build such a dataset named UQ-AAS21. As the first step, we target Amazon Alexa [1], the most popular VPA service at present. We browse through all accessible Alexa applications (or *skills*) in the Amazon Skill store, and download the raw data on the webpage of each skill. We also download all links available on the webpage, so that rigorous information of the skills can be taken into record. This enables our dataset to include useful information that may have overlooked by previous studies, such as the privacy policies and user review comments. Overall, we have included 65,195 skills in our dataset, and comprehensive information about them is incorporated, including invocation names, developers, user reviews, among a total of 16 attributes. All these data are pre-processed and well-formatted to be easily accessible.

We explore the comprehensiveness and usability of our dataset with two preliminary studies. First, we conduct a demographic study on the skills and their developers. Our dataset enables us to reveal several enlightening facts. For example, the analysis of the top popular skills reveals their distribution among the categories. We summarize the quality of a skill from its user rating, and this allows us to explore the potential correlation between a skill’s popularity and its quality. Our preliminary semantic analysis of user reviews, taking two popular skills as case studies, shows users’ negative attitude towards the in-skill purchase. Our statistical analysis of the developers shows that high productivity may not imply high quality. Also, the skills provided by third-party developers could be susceptible to malicious attacks.

Second, we conduct a study on the availability of skills’ privacy policies and terms of use. It reveals that the coverage of developer privacy policy (DPP) and developer terms of use (DTOU) is low. This should raise an alert to all

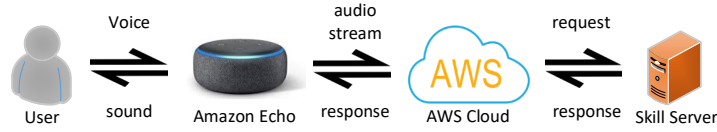


Fig. 1: Interaction flow of an Alexa skill application

stakeholders in the VPA ecosystem, especially in the context that many countries around the world start imposing strict administrative procedures on the user data protection, such as the EU General Data Protection Regulation (GDPR) [5].

Contributions. In summary, this work makes the following contributions.

- **A comprehensive and highly usable dataset.** We build a dataset that includes 65,195 Amazon Alex skills and their rigorous information. It is demonstrated to be highly usable. We have released it to the public to facilitate future research.
- **Two studies based on our dataset.** We conduct two studies on the skills and their developers, revealing some enlightening findings regarding their demography. Our work confirms the findings of previous studies, e.g., Lentzsch et al. [18], and supplements their results with analysis on the data up to May 2021.
- **Advancement of the ecosystem.** Our work provides new perspectives for third-party developers and promotes high-quality skills, contributing to the healthy development of the ecosystem.

2 Background and Related Work

2.1 Background

Amazon Alexa is the most popular VPA service at present. It allows third-party developers to create applications named skills. The interaction flow with a skill is shown in Fig. 1. When there is a voice input from the user, the Alexa Echo device converts it to an audio stream, which is then sent to the Alexa Cloud. The cloud recognizes texts from the audio stream and deliver them to the server of the corresponding skill [11]. All voice recognition and conversion are processed by Alexa Cloud [3] and the skills can only access the processed texts given by Alexa Cloud.

Amazon enforces a vetting process on the skills before they are put onto the skill store. The developer has to test and ensure the usability and ethical policy for their skill to be approved. Amazon Alexa has thousands of skills available on the skill store. However, since they are designated for personal use, security and privacy concerns can be raised. Some studies have reported that several issues exist with Alexa skill vetting process. For example, a dishonest or malicious developer can change the code of the skill server after the skill has been approved

by Amazon, which opens the door for malicious activities [18]. Therefore, in this work, we also include the availability of developer privacy policy and terms of use into our dataset, and show how our dataset can facilitate the research on security and privacy of skills.

2.2 Related Work

Alexa-related Datasets. Lentzsch et al. [18] propose a large-scale data analysis of Amazon Alexa. They collect information from skill store in seven different countries, and their dataset is mainly used to analyze Alexa’s security and privacy issues. Compared with it, our dataset includes updates in 2021, and contains more attributes including application icon, in-skill purchase, supported language and other related information to make the datasets more complete. It also enables some new researches, such as analyzing the relationship between the skills’ popularity and other attributes like names and icons.

Another related dataset is from Gao et al. [14]. It collects 55,502 user reviews of Amazon Echo from 2015 to 2017. These reviews can reflect the attitudes of Alexa’s users about Amazon’s speech recognition capability, but they are not reviews about Alexa skills. In contrast, ours contains the rating of skills, which can be used to analyze the popularity and users’ attitude about the skills.

Virtual Personal Assistants. Gokhan et al. [25] focus on how a VPA can understand the natural languages and distinguish the “target” of voice inputs. They use numerous random web search queries and VPA utterance datasets to train a binary SVM classifier. This study indicates that many VPA-related areas require relevant datasets to support research and train artificial intelligence. This is one of our motivation to build this dataset, which will facilitate more research in this area.

Privacy concerns on Alexa platform. Chung et al. [13] point out if the cloud native data of intelligent virtual assistant (IVA) was leaked, malicious attackers can not only obtain the detailed usage history of IVA services, but also reveal more information about users through data analysis. They collect data covering three months from Alexa cloud, and show that user’s personal information such as interests, and sleeping and wake-up patterns can be identified by analyzing this dataset. This work also shows that the data collection process may contain potential user identity and private information. To avoid collecting any user private information, we only scrape the information that is publicly available and not related to users’ identity.

In addition to the data collection process, some recent studies also show that privacy weakness may arise from other surfaces such as the enforcement of privacy policy [18], the integration of multi-party IoT platforms [21, 26], and the data analysis pipeline [30, 31]. For example, a large-scale analysis report of Alexa skill [18] finds that only 28.5% of skills in the US Amazon store provides a privacy policy. Our work confirms the findings and supplements their results with analysis on the data up to May 2021.

In a study in 2017 [9], researchers have deeply looked into those skills that provided privacy policy links, and find that 9.4% of them cannot be retrieved due

to 404 errors or access denied errors. Additionally, some policy links direct to the developer’s website instead of a specific privacy policy page, which showed the sloppy attitude of some developers. In addition, the relationship between privacy policy statistics and skill’s star rating has also been analyzed. Surprisingly, they find that the skills with the lowest rating have the highest rate of inclusion of a privacy policy (26%). This research gives some inspirations in our data analysis process. We try to visit the privacy policy link and the terms of use link, and find that some links cannot be accessed or are redirected to a wrong page.

When dealing with privacy concerns, there is a study on how to improve user trust for Alexa System [12]. In this work, researchers design a skill application on Alexa platform and conduct a user study. The results show that when users can customize both content and privacy settings, user trust is highest, but may cause reluctance among sophisticated users. In our research, we are inspired from these aspects when collecting and analyzing the data of Amazon Alexa skills. We believe these findings can also have implications for designing more privacy-sensitive and trustworthy VPA apps.

3 The UQ-AAS21 Dataset

The UQ-AAS21 Dataset consists of two datasets, UQ-AAS21-I and UQ-AAS21-II, include rigorous information of Amazon Alexa skills. UQ-AAS21-I provides basic information about skills and UQ-AAS21-II summarizes attributes related to privacy issues. Our work on the UQ-AAS21 dataset includes three phases: data scraping, data processing and data analysis. We discuss on the data scraping and data processing in this section, and defer the data analysis to Section 4. In the data scraping phase, we scrape the web pages of 65,195 Alexa skills, strictly following the ethics of scraping. In the data processing phase, we extract and format the key information.

3.1 Data Scraping

We build a crawler with Python 3.6 to scrape data from Amazon skill store. Before we start crawling, we analyze the target web pages’ source and the architecture of the skill store website, to make sure we cover all the essential information in our crawling. We find that the skills are arranged in 23 categories, such as “Business & Finance” and “Home Services”. The skills are unevenly distributed in these categories. For those categories with more skills, we assign more threads from the maintained thread pool for crawling.

During the crawling process, we use the Requests library [7] to send HTTP requests to the server. To pass the anti-robot checking of Amazon, we register a testing account, and reuse its HTTP headers and session cookies. Our crawler downloads the pages of each category in which the skills are catalogued, and uses BeautifulSoup [2] library to extract the URLs of skills from the pages. The crawler then requests for the skills’ web pages through the URLs, and stores the HTML files.

During our crawling process, sometimes the attempts may trigger Amazon’s anti-crawler mechanisms. Four types of mechanisms may be triggered: (1) The crawler is redirected to the “Robot Check” pages. If it fails to pass the anti-robot checking, the connection will be terminated. (2) When the crawler visits Amazon’s websites too frequently, Amazon might reject the next request, and the crawler exits with an error message of “Connection Reject by Peers”. (3) After the crawler has accessed the website with the same IP address for multiple times, Amazon blocks that IP address for a period ranging from a few hours to a day. (4) After scraping 500-800 web pages, Amazon sends a status code 403 such that the correct information page cannot be visited. To prevent these mechanisms from being triggered frequently, we add a longer random waiting time to each visit interval ranged from 5-10 seconds, and also distribute our crawler across multiple virtual machines in our server which has multiple IP addresses.

Ethical Considerations. To avoid introducing negative influence to the Amazon website and its users, we strictly follow the ethics of scraping. We comply with Amazon’s terms of use policy before attempting website crawling, and also honour the “Robot.txt” to avoid scraping sensitive user information. All crawled data is public available and used for legitimate purposes, without any user private information or secret information of Amazon. In addition, as mentioned above, we control the frequency of scraping and make the program run at a low speed to ensure that the availability of Amazon servers would not be affected.

3.2 Data Processing

After scraping the web pages of skills, we start our data processing to extract crucial information of them. Meanwhile, we also organize the information to make our dataset easy to use. First, we analyze the page structure of the downloaded HTML file. we use BeautifulSoup library to extract critical information from the HTML files, including skill name, developer invocation name, number of rating, etc. After that, we store the information of each skill in a row of a table. Skills from the same category are stored in the same table.

In the next step, we fix the error data. We mainly handle the wrong pages or the 404 error pages in this step. Wrong pages are extracted because we scrape Amazon’s advertisement pages unexpectedly. The 404 error pages or Amazon’s robot check pages exist since the Amazon’s anti-scraping mechanisms may be triggered sometimes. When this happens, the corresponding skill page information is lost. To make the checking efficient, we check the size of the files, as the size of normal skill pages ranges from 200-400kb, and the size of error pages and 404 pages are less than 100 kb. If any of such pages is found, we re-scrape it.

The BeautifulSoup library is used to parse the structure of the HTML. The data extraction is also carried out per categories. We create an Excel file for each of the 23 categories and write information into the corresponding columns. In this way, the data can be easily converted into data frames, e.g., using the Pandas library, to fit into the desired input of various statistical operations.

Table 1: Skill feature overview for UQ-AAS21-I dataset

#	attributes	Description	Example	Note
1	Skill Name	The Alexa skill name	“Deadpool”	
2	Path	The path to the corresponding skill web page	“https://www.amazon.com/Austin-Koontz-Deadpool/dp/B0754V RGVX”	
3	Developer	The developer of this skill	“by Austin Koontz”	The developer could be both Amazon and third-party developers
4	Supported Languages	Different languages that this skill can support	“English (US) English (CA)”	Alexa skills can support 8 different languages ¹
5	Average Rating	The average rating score of this skill (out of 5)	“2.9 out of 5”	
6	Number of Rating	The number of ratings that this skill receives	“340”	
7	Price	The purchase price of this skill	“Free to Enable”	Label “Free to enable” if the skill is free
8	In-skill Purchase	The price of in-skill purchase of this skill	“Premium \$9.99/month”	Empty if the skill doesn’t have in-skill purchase
9	Wake-up Words	The corresponding words that can wake up Amazon Echo and open this skill by voice	“Alexa, open deadpool”	Each skill has 1 to 3 wake-up words
10	Icon	The icon of this skill	“https://images-na.ssl-images-amazon.com/images/I/717ACt+384L.png”	Store as a link to a picture
11	Category	The corresponding category that this skill belongs to	“Movies & TV”	The Amazon Alexa skills system has overall 23 categories

3.3 Dataset Features

After the data processing phase, UQ-AAS21 dataset is generated and divided into two parts. UQ-AAS21-I dataset includes 11 attributes, while UQ-AAS21-II dataset include 6 attributes, which covers all 16 attributes of the skill (the skill name appears in both datasets). These attributes together describe the feature of a skill. The specific attributes are listed and explained in Table 1 and 2. We separate some attributes into UQ-AAS21-II dataset because after comparing the related researches about Amazon Alexa field, we find that the privacy issues of Amazon Alexa platform is important. Therefore, we decide to put the privacy-related attributes into a single dataset.

Some attributes in UQ-AAS21-I and UQ-AAS21-II datasets have multiple possible values. For example, the “Supported Languages” attribute in UQ-AAS21-II dataset indicates the languages supported by the skill. As shown in Table 1, Alexa skills can support 8 different languages at most, where each language

¹ The supported languages include English, German, Italian, Spanish, Portuguese, Japanese, French, Hindi, where each language can also correspond to different country codes.

Table 2: Skill feature overview for UQ-AAS21-II dataset

#	attributes	Description	Example	Note
1	Skill Path	The link to skill application	See in Table 1	
2	Access	The permissions type that this skill request	“Email Address, Reminders”	Label “No permission needed” if the skill requests no permission; The permission information has 12 subtypes ²
3	Invocation Name	The unique invocation name of this skill	“money exchange”	Label “Not Exists” if the skill has no invocation name
4	Description	The description of the skill (aims, usage, related information, etc.)	“This is your local Flash Briefing ...” ³	
5	Developer Privacy Policy	The Developer Privacy Policy of this skill	“https://www.nbcuniversal.com/privacy”	Label “Not Exists” if the skill has no developer privacy policy
6	Developer Terms of Use Document	The Developer Terms of Use Document of this skill	“https://www.cnn.com/cnn-amazon-alexa-skills-terms-of-use/”	Label “Not Exists” if the skill has no developer terms of use

can also correspond to a different country code. The “Access” attribute in UQ-AAS21-II dataset contains what user information is accessed by the skill. The skill needs users’ permission to access the listed information in this field. It can be categorized into 12 types of Alexa skill permissions (listed in Table 2). We also include the “application icon” attribute to facilitate the research that may involve images. For example, the graphics and colors of the Alexa skill applications’ icons can be analyzed to find out whether they are related to the popularity of the applications, so as to provide insights for Amazon Alexa developers in designing their icons.

4 Preliminary studies based on UQ-AAS21 datasets

We conduct two studies based on the UQ-AAS21 dataset to explore its usability.

4.1 Demographic study

In the demographic study, we examine the basic information about the collected skills and their developers.

² The subtypes of Access: Device Address, Device Country and Postal Code, Email Address, Alexa Notifications, Location Services, Mobile Number, Reminders, Lists Read Access, Lists Write Access, First Name, Full Name, Amazon Pay.

³ Full Text: “This is your local Flash Briefing to help you sell, buy, and rent Southlake real estate with your Southlake real estate agent SONYA KENNEDY.”

Table 3: The number of skills in each category

Categories	Number of skills	Categories	Number of skills
Business & Finance	3,824	Movies & TV	949
Communication	567	Music & Audio	6,396
Weather	846	News	6,396
Connected Car	172	Novelty & Humor	3,472
Education & Reference	6,414	Productivity	5,206
Food & Drink	1,336	Smart Home	2,817
Games & Trivia	6,383	Shopping	408
Health & Fitness	2,214	Social	1,881
Home Services	343	Sports	1,733
Kids	3,759	Travel & Transportation	1,197
Lifestyle	6,204	Utilities	1,432
Local	1,246	Total	65,195

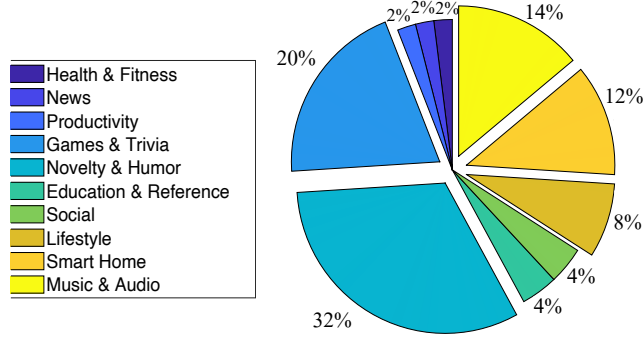


Fig. 2: Category distribution of top 50 popular skills

4.1.1 Skills analysis

We reuse the 23 categories of Alexa skill store to organize the collected 65,195 skills. Their distribution among the categories is listed in Table 3. The smallest category is “connected car”, which take up 0.26% of total skills, while the largest is “education and reference”, which occupied 9.84% of total skills.

Popular skills. We first analyze the top 50 popular skills⁴. Fig. 2 shows their distribution among the categories. Most of them (32%) fall into the “Novelty & Humor” category, while the second popular category is “Games & Trivia”, with 20% of the top 50 skills. The “Smart Home” and “Music & Audio” categories have the similar popularity, which has 12% and 14% of the skills respectively. The “Health & Fitness”, “News” and “Productivity” have the least skills, which is 2% of the total.

⁴ The popularity of a skill is measured based on the number of ratings it receives.

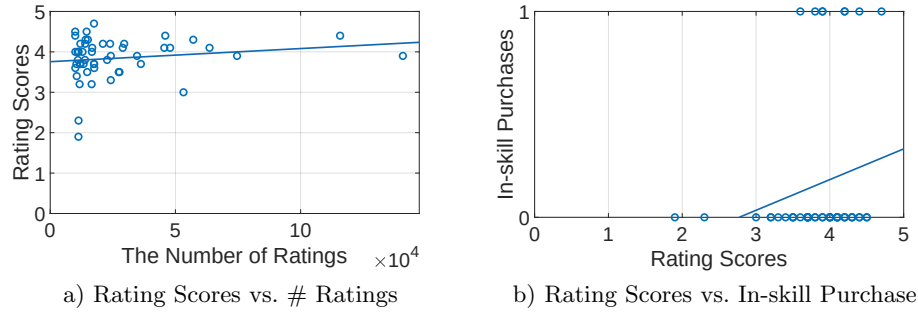


Fig. 3: Analysis of top 50 popular skills

Table 4: Top ten popular skills

#	Top Ten Skills with the Most of Rating	The Number of Ratings	Average Score	Price	In-skill Purchase	Developer
1	4AFart	140,832	3.9 out of 5	Free to Enable		by Bloatware
2	Big Fart	115,704	4.4 out of 5	Free to Enable	The Extreme Farts Extension Pack \$1.99	by HUGO.FM
3	Jurassic Bark	74,554	3.9 out of 5	Free to Enable		by Amaury Gallais
4	Samuel L.Jackson - celebrity voice for Alexa	73,455	4.1 out of 5	\$4.99		by Amazon
5	Find My Phone	57,082	4.4 out of 5	Free to Enable		by Matchbox.io Inc
6	Jeopardy!	53,326	3.3 out of 5	Free to Enable		by Sony Pictures Television
7	Pikachu Talk	47,865	4.1 out of 5	Free to Enable		by The Pokemon Company
8	Happy Birthday	45,941	4.4 out of 5	Free to Enable	Premium Birthday Bundle \$1.99	by HUGO.FM
9	iRobot Home	45,532	4.1 out of 5	Free to Enable		by iRobot
10	Repeat After Me	36,191	3.7 out of 5	Free to Enable		by Spraypaint

In Fig. 3, we demonstrate two observations on the rating scores of top 50 popular skills. (1) *Rating scores vs. the number of ratings*. As shown in Fig. 3a, the rating score of an app slightly increases with the number of ratings it receives, indicating a positive correlation between a skill’s popularity and quality. (2) *Rating scores vs. in-skill purchase*. The skills with in-skill purchase achieves slightly higher ratings scores than those without (Fig. 3b). This may be attributed to its positive affect on users’ engagement and retention rates. For example, the users are more likely to make an in-app purchase after playing the game for a while.

In Table 4, we list the top 10 popular skills in detail. As expected, the majority of popular skills are free to enable. Also, skills provided by reputable companies, such as Microsoft and Amazon, usually achieve high popularity as well as high rating scores.

Semantic analysis. We conduct a preliminary semantic analysis of user reviews. Fig. 4 show the keyword extracted from the reviews of two skills from the top

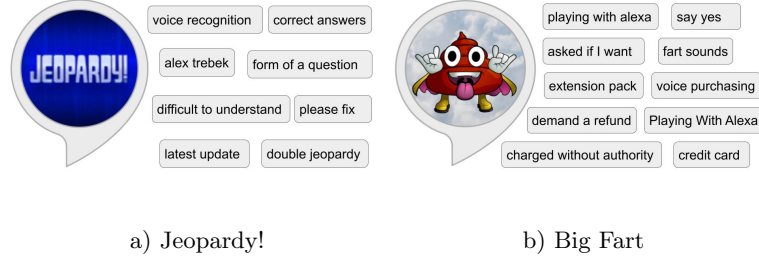


Fig. 4: Keywords extract from skills' reviews

ten list, namely “Jeopardy!” and “Big Fart”. For “Jeopardy!”, users’ opinions tend to be negative, as shown in Fig. 4a the extracted keywords such as “difficult to understand” and “please fix”. This is also reflected in the app’s user rating score (3.3/5), which is the lowest of the 10 most popular skills. For “Big Fart”, although it achieves high rating (4.4/5), our semantic analysis shows that users have negative opinions on its in-skill purchase (Fig. 4b).

4.1.2 Developer analysis

We also analyze the top developers who have provided the most skills. In Table 5, we list the top ten developers. We find that high productivity may not imply high quality. For example, in the skills provided by the developer “My Home Agent” (the second productive developer), it contains a link which even directs to a 404-error page. In addition, the utilization rate of some skills provided by top developers is relatively low. Taking the top one productive developer as an example, among the 1,445 skills, only 15 skills (1.03%) have user rating.

Notably, some skills provided by extraordinarily productive developers may be susceptible to malicious injection. For example, the top developer “InfoByVoice” (who has provided 1,455 skills) put in an advertisement link in their skills’ description page, which directs to a website that provides skill application templates and allows anyone to create skills. This makes it possible for a malicious user to publish a skill under the company name, and such process may raise security concerns.

4.2 Analysis of privacy policy and term of use document

We find that the coverage of the developer privacy policy (DPP) and the developer terms of use (DTOU) for Alexa skills is low. Among the 65,195 skills, the number of skills that fail to provide a DPP is higher than two-thirds of the total number (67.7%), and the number of skills that fail to provide a DTOU is even higher (77.7%). Table 6 shows the ratios of skills without DPP or DTOU over

Table 5: Top ten developers with the most of skills

#	Top Developers with the Most Skills	Skill Counts	#	Top Developers with the Most Skills	Skill Counts
1	InfoByVoice	1455	6	SnoCountry	445
2	My Home Agent	693	7	GateHouse Media, LLC	431
3	Patch.com	489	8	sachin nelwade	352
4	Online Radio	474	9	Amazon	340
5	GoVocal.AI	449	10	VoiceSkillsInc	320

all skills in the 23 categories. The “Smart Home” category performs the best, with only 6% of skills that do not provide a DPP, while the “Kids” category has the highest percentage of missing DPP of 87.6%.

5 Potential Usage of UQ-AAS21

In this section, we outline a few potential usages of the UQ-AAS21 dataset, surrounding the healthy development of the skill ecosystem.

Security issue identification. We see the opportunities of UQ-AAS21 dataset contributing to the research area in identifying security issues [26]. The rigorous information (such as links, invocation words, access permission) provided by the dataset can be used to identify potential flaws, which may include loopholes that can be exploited by a malicious actor to publish fictitious skills, adversarial examples that can trick voice-based interfaces, and various sophisticated attacks against VPA such as voice-squatting attacks and hidden voice attacks [17, 19, 27].

Kumar et al. [17] conduct a Speech-to-Text test on Alexa. They create a skill and use Alexa as a transcription service. In the testing, they record the voice inputs, text outputs and recognized query commands. Then, they present a skill squatting attacks that aim to construct a malicious skill and route users to it by a systematic error in the interpretation of the input. The “Wake-Up Words” attributes in our dataset (see Table 1) can help researchers to understand the common structure of skills’ voice commands and also can be used to compare to the swapped “malicious” wake-up words.

Privacy policy compliance. Alexa mandates a privacy policy link for skills that request certain permission APIs. The DPP and DTOU information provided by the UQ-AAS21 dataset can be used to investigate privacy issues of IoT ecosystems [21] from the perspectives of (1) *the effectiveness of the privacy policy*, e.g., whether privacy policy links actually serve their purpose of informing users of their data practices; (2) *the compliance of the privacy policy*, i.e., whether the privacy violations (such as access policy breaches, unintended flows, over-privilege access) may arise throughout the lifecycle of the interaction flow.

For example, the attribute “Access” in UQ-AAS21 dataset can be used in privacy compliance analysis. The skills can be configured to request permission to access user’s personal information (there are 12 subtypes of permissions in total, see Table 2). Lentzsch et al. [18] report instances where skills bypass the

Table 6: Ratios of skills without developer privacy policy (DPP) or developer terms of use(DTOU)

Categories	Ratio Without DPP	Ratio Without DTOU	Categories	Ratio Without DPP	Ratio Without DTOU
Business & Finance	2280/3824 (59.6%)	2691/3824 (70.4%)	Movies & TV	780/949 (82.1%)	827/949 (87.1%)
Communication	218/567 (38.4%)	267/567 (47.0%)	Music & Audio	3316/6396 (51.8%)	3776/6396 (59.0%)
Weather	693/846 (81.8%)	762/846 (90.0%)	News	3942/6396 (61.6%)	4930/6396 (77.1%)
Connected Car	42/172 (24.3%)	109/172 (63.0%)	Novelty & Humor	3021/3472 (87.0%)	3252/3472 (93.6%)
Education & Reference	4993/6414 (77.8%)	5322/6414 (83.0%)	Productivity	3597/5206 (69.1%)	4200/5206 (80.7%)
Food & Drink	992/1336 (74.2%)	1121/1336 (83.8%)	Smart Home	170/2817 (6.0%)	1620/2817 (57.5%)
Games & Trivia	4931/6383 (77.2%)	5200/6383 (81.5%)	Shopping	142/408 (34.7%)	203/408 (49.6%)
Health & Fitness	1263/2214 (57.0%)	1572/2214 (71.0%)	Social	1335/1881 (70.9%)	1471/1881 (78.2%)
Home Services	142/343 (41.3%)	205/343 (59.6%)	Sports	1340/1733 (77.3%)	1463/1733 (84.4%)
Kids	3293/3759 (87.6%)	3408/3759 (90.6%)	Travel & Transportation	973/1197 (81.2%)	1082/1197 (90.3%)
Lifestyle	4513/6204 (72.7%)	4854/6204 (78.2%)	Utilities	1222/1432 (85.3%)	1283/1432 (89.5%)
Local	933/1246 (74.8%)	1049/1246 (84.1%)	Total	44131/65195 (67.7%)	50667/65195 (77.7%)

permission APIs and directly request information from users. In this case, user’s privacy information is collected and used without authorization. To identify such cases, researchers can record the personal information acquired by the skills and compare them with their declared permissions in our dataset.

Skill quality improvement. The user reviews provided by the UQ-AAS21 dataset can guide third-party developers with new perspectives and promote high-quality skills. It can facilitate the relevant research areas such as semantic analysis of VPA ecosystem, user profiling, and recommender systems. In addition, the UQ-AAS21 dataset can also contribute to the study of multiple levels of linguistic and paralinguistic features [28] and facilitate the research in the voice recognition area.

6 Conclusion

In this work, we build a comprehensive dataset of Amazon Alexa skills for research use. It contains 65,195 skill applications and their rigorous information including the privacy policy, user reviews, among over all 16 attributes. We also demonstrate the comprehensiveness and usability of our dataset with two preliminary studies. In the demographic study, we reveal the positive correlation between a skill’s popularity and its quality. Our semantic analysis of user reviews

shows users' negative attitude towards the in-skill purchase. Our statistical analysis of the developers shows that high productivity may not imply high quality, and we also identify potential security issues in a developer's development model. In the privacy analysis, we review the coverage of developer privacy policy and developer terms of use of Alexa skills is low. We hope our dataset will facilitate the future research in the health of various VPA ecosystems.

References

1. Amazon alexa voice ai — alexa developer official site (2021), <https://developer.amazon.com/en-US/alexa>
2. Beautiful soup documentation — beautiful soup 4.9.0 documentation (2021), <https://www.crummy.com/software/BeautifulSoup/bs4/doc/>
3. Create alexa skills kit — custom voice model skills (2021), <https://developer.amazon.com/en-US/docs/alexa/custom-skills/understanding-how-users-invoke-custom-skills.html>
4. Facts and statistics about virtual assistants (2021), <https://www.statista.com/topics/5572/virtual-assistants/>
5. General data protection regulation (2021), <https://gdpr-info.eu>
6. Google play store (2021), <https://play.google.com/store?hl=en&gl=US>
7. Requests: Http for humans™ — requests 2.26.0 documentation (2021), <https://docs.python-requests.org/en/master/>
8. Total number of amazon alexa skills from january 2016 to september 2019 (2021), <https://www.statista.com/statistics/912856/amazon-alexa-skills-growth/>
9. Alhadlaq, A., Tang, J., Almaymoni, M., Korolova, A.: Privacy in the amazon alexa skills ecosystem. *Star* **217**(11) (1902)
10. Ammari, T., Kaye, J., Tsai, J.Y., Bentley, F.: Music, search, and iot: How people (really) use voice assistants. *ACM Trans. Comput. Hum. Interact.* **26**(3), 17–1 (2019)
11. Bhargava, M., Safari, a.O.M.C.: Alexa skills projects : build exciting projects with Amazon Alexa and integrate it with Internet of Things. Birmingham, England ; Mumbai : Packt, 1st edition edn. (2018), <https://ebookcentral.proquest.com/lib/canterbury/detail.action?docID=5446037>, electronic reproduction. Boston, MA : Safari. Available via World Wide Web., 2018
12. Cho, E., Sundar, S.S., Abdullah, S., Motalebi, N.: Will deleting history make alexa more trustworthy? effects of privacy and content customization on user experience of smart speakers. In: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. pp. 1–13 (2020)
13. Chung, H., Lee, S.: Intelligent virtual assistant knows your life. *arXiv preprint arXiv:1803.00466* (2018)
14. Gao, Y., Pan, Z., Wang, H., Chen, G.: Alexa, my love: Analyzing reviews of amazon echo. In: *2018 IEEE SmartWorld, Ubiquitous Intelligence Computing, Advanced Trusted Computing, Scalable Computing Communications, Cloud Big Data Computing, Internet of People and Smart City Innovation (SmartWorld/SCALCOM/UIC/ATC/CBDCom/IOP/SCI)*. pp. 372–380 (2018). <https://doi.org/10.1109/SmartWorld.2018.00094>
15. Guo, Z., Lin, Z., Li, P., Chen, K.: Skillexplore: Understanding the behavior of skills in large scale. In: *29th {USENIX} Security Symposium ({USENIX} Security 20)*. pp. 2649–2666 (2020)

16. Kinsella, B.: Google assistant actions total 4,253 in january 2019, up 2.5 x in past year but 7.5% the total number alexa skills in us. Voicebot. ai, February **15** (2019)
17. Kumar, D., Paccagnella, R., Murley, P., Hennenfent, E., Mason, J., Bates, A., Bailey, M.: Skill squatting attacks on amazon alexa. In: 27th {USENIX} Security Symposium ({USENIX} Security 18). pp. 33–47 (2018)
18. Lentzsch, C., Shah, S.J., Andow, B., Degeling, M., Das, A., Enck, W.: Hey alexa, is this skill safe?: Taking a closer look at the alexa skill ecosystem. In: 28th Annual Network and Distributed System Security Symposium (NDSS 2021). The Internet Society (2021)
19. Lit, Y., Kim, S., Sy, E.: A survey on amazon alexa attack surfaces. In: 2021 IEEE 18th Annual Consumer Communications Networking Conference (CCNC). pp. 1–7 (2021). <https://doi.org/10.1109/CCNC49032.2021.9369553>
20. Lopatovska, I., Rink, K., Knight, I., Raines, K., Cosenza, K., Williams, H., Sorsche, P., Hirsch, D., Li, Q., Martinez, A.: Talk to me: Exploring user interactions with the amazon alexa. *Journal of Librarianship and Information Science* **51**(4), 984–997 (2019)
21. Mahadewa, K., Zhang, Y., Bai, G., Bu, L., Zuo, Z., Fernando, D., Liang, Z., Dong, J.S.: Identifying privacy weaknesses from multi-party trigger-action integration platforms. In: Proceedings of the 30th ACM SIGSOFT International Symposium on Software Testing and Analysis. pp. 2–15 (2021)
22. Malkin, N., Deatrick, J., Tong, A., Wijesekera, P., Egelman, S., Wagner, D.: Privacy attitudes of smart speaker users. *Proceedings on Privacy Enhancing Technologies* **2019**(4) (2019)
23. Schönherr, L., Kohls, K., Zeiler, S., Holz, T., Kolossa, D.: Adversarial attacks against automatic speech recognition systems via psychoacoustic hiding. arXiv preprint arXiv:1808.05665 (2018)
24. Sciuto, A., Saini, A., Forlizzi, J., Hong, J.I.: ” hey alexa, what’s up?” a mixed-methods studies of in-home conversational agent usage. In: Proceedings of the 2018 Designing Interactive Systems Conference. pp. 857–868 (2018)
25. Tur, G., Deoras, A., Hakkani-Tür, D.: Detecting out-of-domain utterances addressed to a virtual personal assistant. In: Fifteenth Annual Conference of the International Speech Communication Association (2014)
26. Wang, K., Zhang, J., Bai, G., Ko, R., Dong, J.S.: It’s not just the site, it’s the contents: Intra-domain fingerprinting social media websites through cdn bursts. In: Proceedings of the Web Conference 2021. pp. 2142–2153 (2021)
27. Yuan, X., Chen, Y., Wang, A., Chen, K., Zhang, S., Huang, H., Molloy, I.M.: All your alexa are belong to us: A remote voice control attack against echo. In: 2018 IEEE Global Communications Conference (GLOBECOM). pp. 1–6 (2018). <https://doi.org/10.1109/GLOCOM.2018.8647762>
28. Zarate, J.M., Tian, X., Woods, K.J., Poeppel, D.: Multiple levels of linguistic and paralinguistic features contribute to voice recognition. *Scientific reports* **5**(1), 1–9 (2015)
29. Zhang, N., Mi, X., Feng, X., Wang, X., Tian, Y., Qian, F.: Dangerous skills: Understanding and mitigating security risks of voice-controlled third-party functions on virtual personal assistant systems. In: 2019 IEEE Symposium on Security and Privacy (SP). pp. 1381–1396. IEEE (2019)
30. Zhang, Y., Bai, G., Li, X., Curtis, C., Chen, C., Ko, R.K.: PrivColl: Practical privacy-preserving collaborative machine learning. In: European Symposium on Research in Computer Security. pp. 399–418. Springer (2020)
31. Zhang, Y., Bai, G., Zhong, M., Li, X., Ko, R.: Differentially private collaborative coupling learning for recommender systems. *IEEE Intelligent Systems* (2020)